

Little Man Computer: Max Find

IT110 Lab Exercise

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The von Neumann Architecture

The Little Man Computer is a pedagogical proxy of a basic von Neumann Architecture.

Key Components:

- **Input/Output:** One channel for each.
- **RAM (Mailboxes):** Stores both data and instructions!
- **CPU:** Contains the Accumulator and Program Counter.

Today's Goal: Find the Maximum

The Problem: How does a computer know which of two numbers is bigger if it can only do basic math?

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The Solution: Subtraction!

- Input A
- Input B
- Calculate $B - A$
- **If the result is positive (≥ 0):** B is the maximum.
- **If the result is negative (< 0):** A is the maximum.

The Assembly Code

Enter this exactly into the Assembly Language Code box:

```
00 INP          // Input A
01 STO 90       // Store A in box 90
02 INP          // Input B
03 STO 91       // Store B in box 91
04 SUB 90       // Accumulator is now (B - A)
05 BRP 10       // If B >= A, jump to line 10
06 LDA 90       // Else (A > B), load A
07 BRA 11       // Jump to Output (skip line 10)
08             // (Empty lines usually ignored)
09             //
10 LDA 91       // Load B
11 OUT          // Output the Accumulator
12 HLT         // Halt
```

Understanding the Branching

Conditional Branching (BRP)

BRP 10 acts like an **IF** statement.

If Accumulator ≥ 0 , go to mailbox 10.

Unconditional Branching (BRA)

BRA 11 acts to skip the **ELSE** block.

We loaded *A* (the max), so we must skip over the instruction that loads *B*.

Time to Code!

Open peterhigginson.co.uk/LMC